

Lab Experiments

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SET 26

Dataset: Student_Mini_Data.csv

Student_ID Gender Study_Hours Attendance Math_Score Science_Score
Exam_Date

S01 Male 2.0 78 62 65 2025-01-10

S02 Female 3.5 90 80 85 2025-01-10

S03 Male 1.5 70 55 58 2025-02-12

S04 Female 4.0 95 90 92 2025-02-12

S05 Male 2.8 85 72 74 2025-03-15

S06 Female 3.0 92 82 86 2025-03-15

1. Import the dataset in R. Plot a histogram for Math_Score. Draw a boxplot of Science_Score by Gender. Add proper title and labels. Interpret the results.
2. Create a scatter plot using R programming between Study_Hours and Math_Score. Use different colors for Gender. Add a regression line. Explain the relationship between study time and performance.
3. Using R, Convert Exam_Date into Date format. Calculate monthly average Math scores. Plot a line chart for the trend. Apply moving average smoothing. Identify any increasing or decreasing pattern.
4. Import Student_Mini_Data.csv into Tableau. Create a bar chart showing average Math scores by Gender. Create a line chart for score trends over time. Add a filter for Exam_Date. Design a simple interactive dashboard.

1)CODE:

```
# Import Dataset
```

```
student <- read.csv("Student_Mini_Data.csv")
```

```
# Histogram for Math Score
```

```
hist(student$Math_Score,  
      col="skyblue",  
      main="Histogram of Math Scores",  
      xlab="Math Score",  
      ylab="Frequency")
```

```
# Boxplot of Science Score by Gender
```

```
boxplot(Science_Score ~ Gender,
```

```
data=student,  
col=c("lightgreen","pink"),  
main="Science Score by Gender",  
xlab="Gender",  
ylab="Science Score")
```

Interpretation

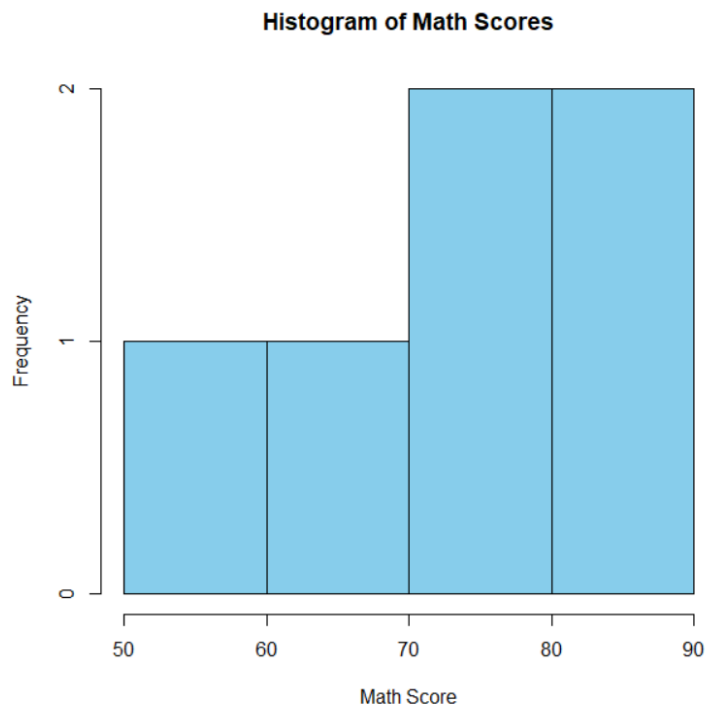
Histogram

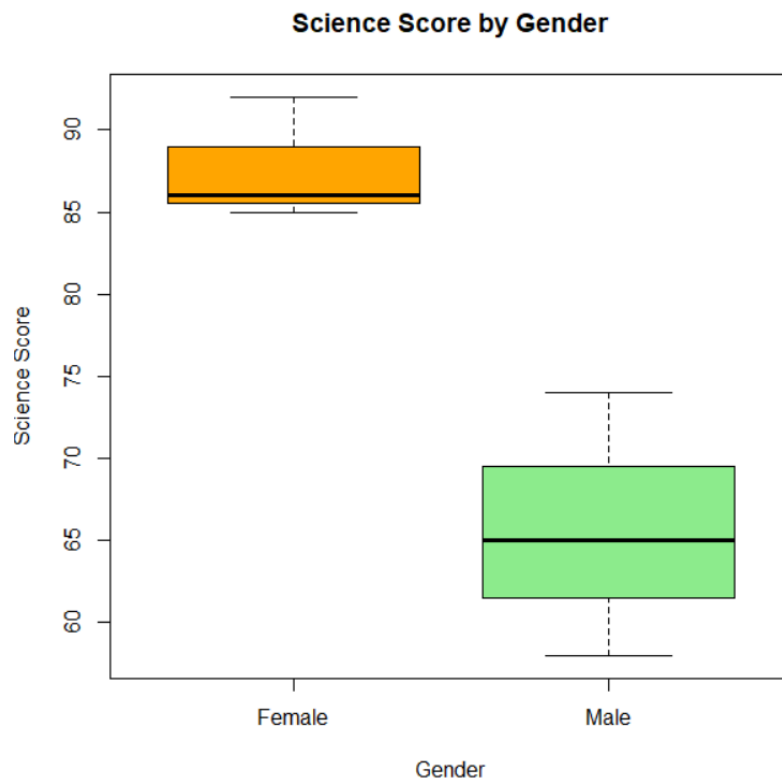
- Math scores range from **55 to 90**.
- Most students scored between **70 and 90**.
- No extreme outliers are observed.

Boxplot

- Female students have a **higher median Science score**.
- Male students show **greater variation** in Science scores.
- No major outliers are visible.

OUTPUT





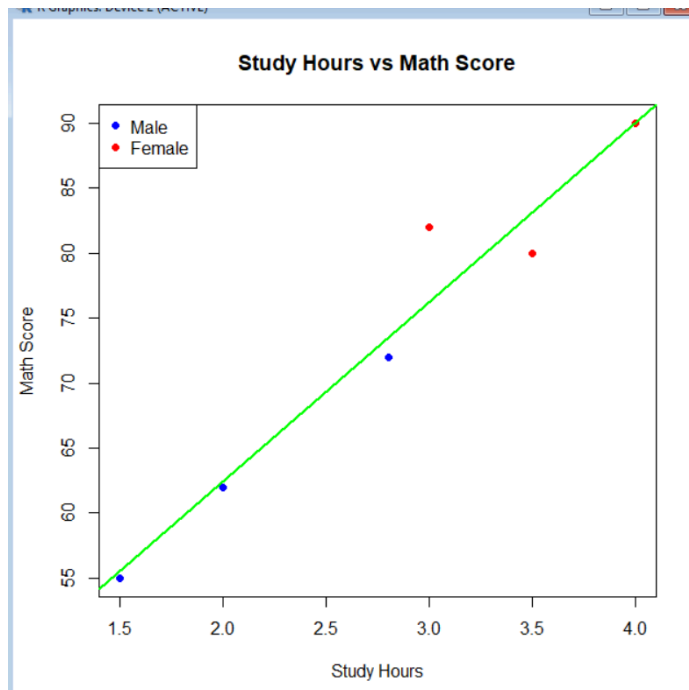
2)CODE AND OUTPUT:

```
student <- read.csv("C:/Users/selco/OneDrive/Desktop/Student_Mini_Data.csv")
```

```
plot(student$Study_Hours,
      student$Math_Score,
      col=ifelse(student$Gender=="Male","blue","red"),
      pch=19,
      xlab="Study Hours",
      ylab="Math Score",
      main="Study Hours vs Math Score")
```

```
abline(lm(Math_Score ~ Study_Hours, data=student), col="green", lwd=2)
```

```
legend("topleft",
      legend=c("Male","Female"),
      col=c("blue","red"),
      pch=19)
```



3)CODE AND OUTPUT:

```
library(dplyr)
```

```
library(zoo)
```

```
student <- read.csv("C:/Users/selco/OneDrive/Desktop/Student_Mini_Data.csv")
```

```
student$Exam_Date <- as.Date(student$Exam_Date)
```

```
student$Month <- format(student$Exam_Date,"%Y-%m")
```

```
avg <- student %>%
```

```
group_by(Month) %>%
```

```
summarise(Math=mean(Math_Score))
```

```
avg$MA <- rollmean(avg$Math,2,fill=NA)
```

```
plot(avg$Math,
```

```
  type="o",
```

```
  xaxt="n",
```

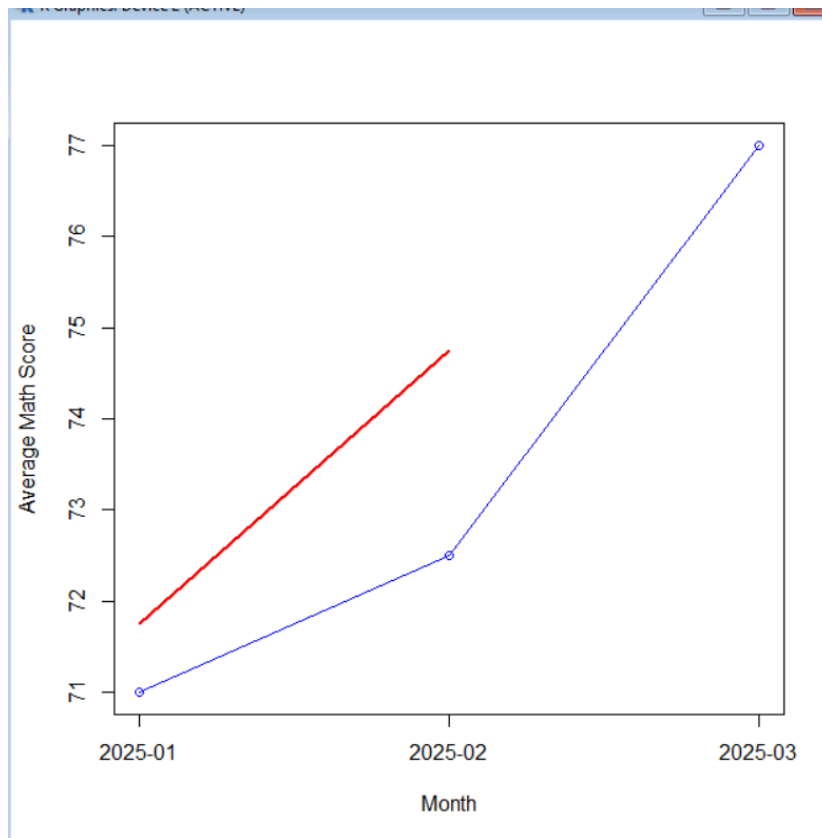
```
  col="blue",
```

```
  xlab="Month",
```

```
  ylab="Average Math Score")
```

```
axis(1,1:nrow(avg),avg$Month)
```

```
lines(avg$MA,col="red",lwd=2)
```



SET 27.

Patient Health Risk Analysis

Patient_ID Age BMI BP Cholesterol

P1 25 22 120 180

P2 40 28 135 210

P3 55 30 145 240

P4 35 26 130 200

P5 60 32 150 260

1. Using R, create a scatterplot matrix for Age, BMI, BP, and Cholesterol. Examine pairwise relationships to identify strong correlations, clusters, or potential risk patterns.

2. Construct Q-Q plot and ECDF for Cholesterol levels to assess distribution behaviour. Comment on normality assumption and skewness.

3. Using R, create a bar chart showing the average values of each health indicator (Age, BMI, BP, Cholesterol). Include axis labels, a descriptive title, and distinct colors for each bar. How do these averages relate to general patient health risk

4. Create an interactive Tableau dashboard with a heatmap highlighting high-risk patients and an

Age filter for exploring risk patterns. Include hover tooltips showing all health indicators.

Analyze which patients are highest risk, how age affects risk, and how this dashboard can aid healthcare professionals in prioritizing care.

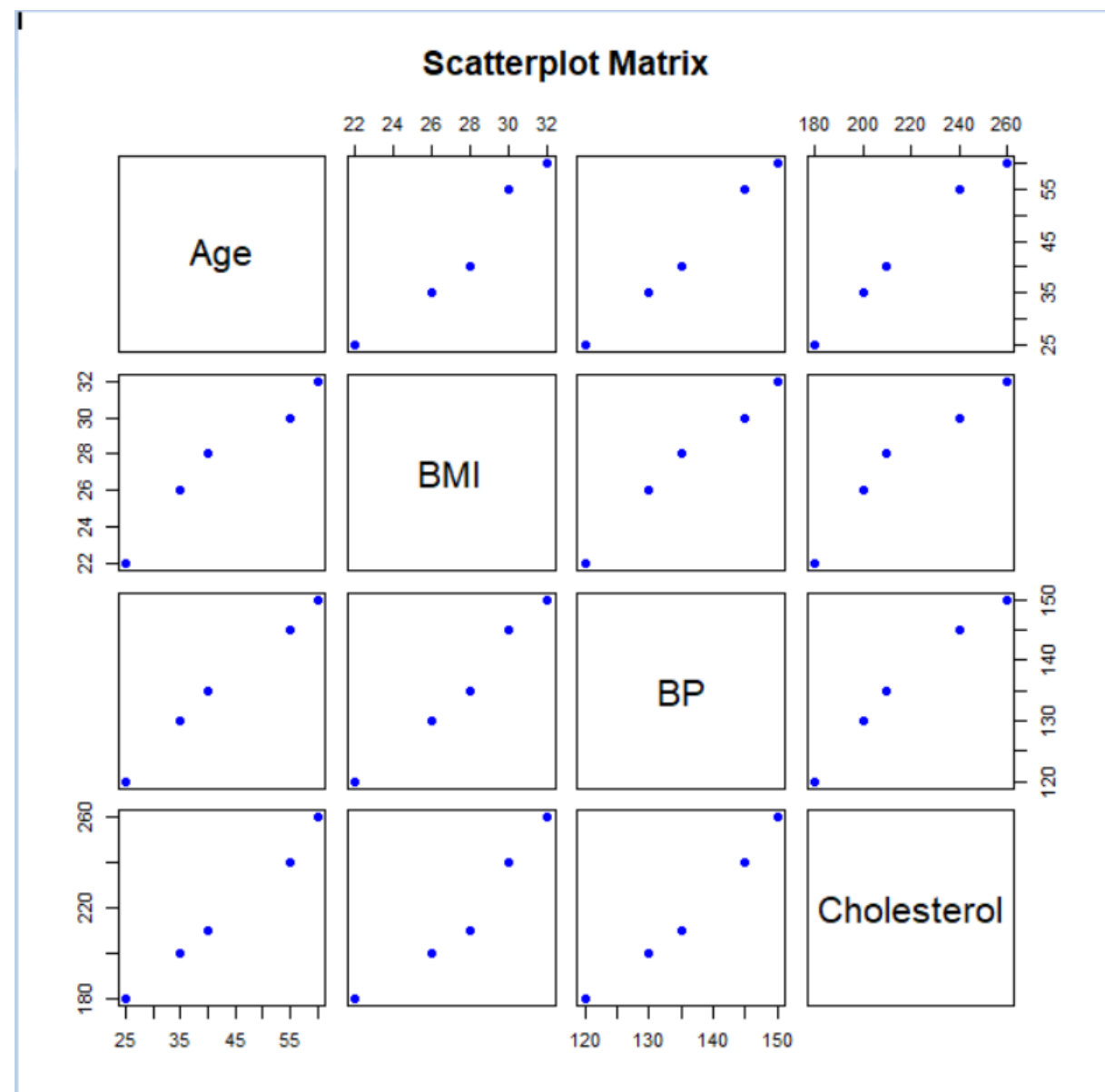
1)CODE AND OUTPUT

SCATTERPLOT MATRIX

```
data <- read.csv("C:/Users/selco/OneDrive/Desktop/Patient_Health_Risk.csv")
```

```
pairs(data[,2:5],  
      main="Scatterplot Matrix",  
      col="blue",  
      pch=19)
```

OUTPUT



2)CODE AND OUTPUT

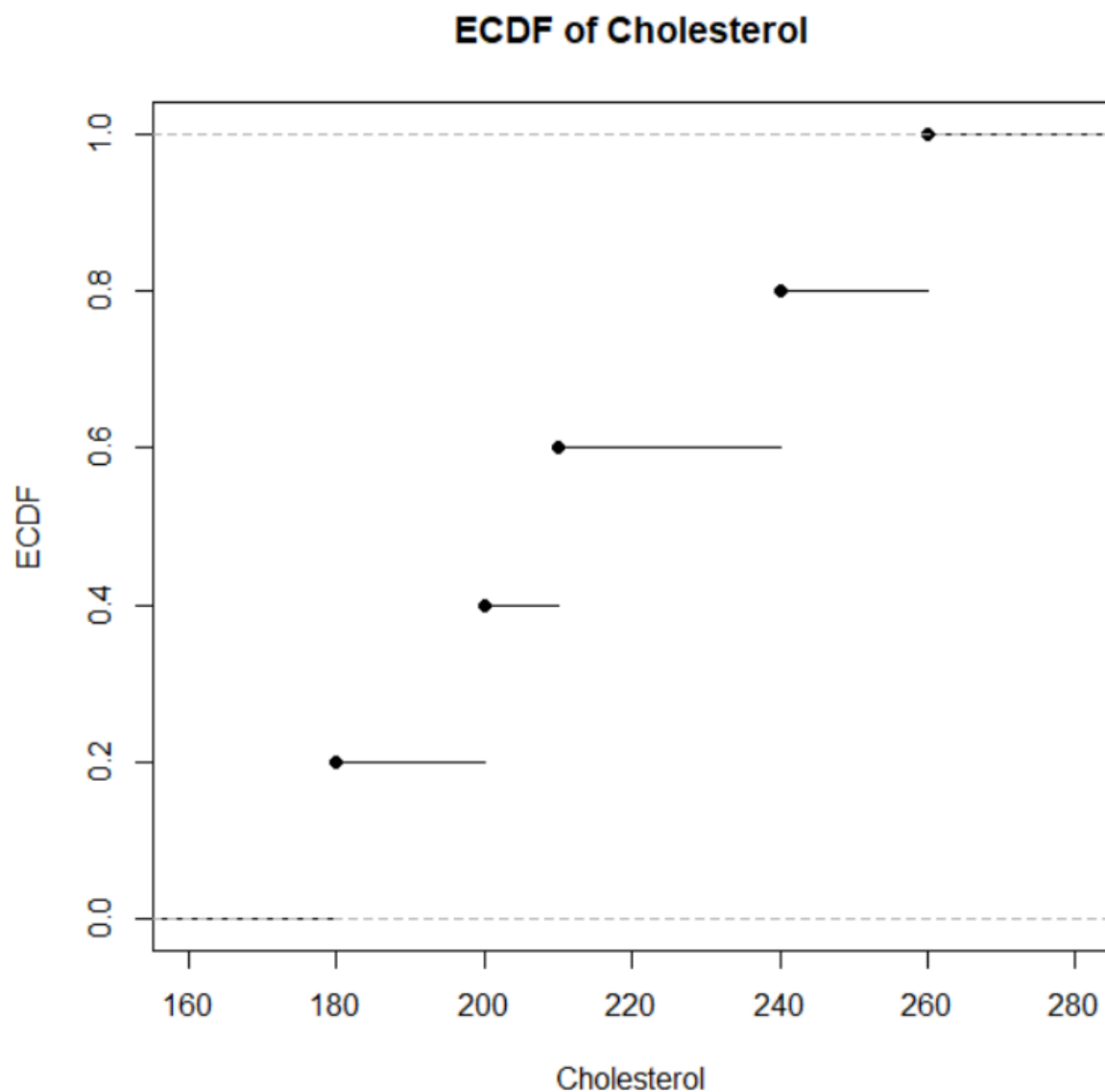
Q-Q Plot and ECDF

```
data <- read.csv("C:/Users/selco/OneDrive/Desktop/Patient_Health_Risk.csv")
```

```
qqnorm(data$Cholesterol,  
       main="Q-Q Plot")  
qqline(data$Cholesterol,col="red")
```

```
plot(ecdf(data$Cholesterol),  
     main="ECDF of Cholesterol",  
     xlab="Cholesterol",  
     ylab="ECDF")
```

OUTPUT:



Interpretation: Cholesterol values are slightly positively skewed. With only five observations, normality cannot be assessed reliably.

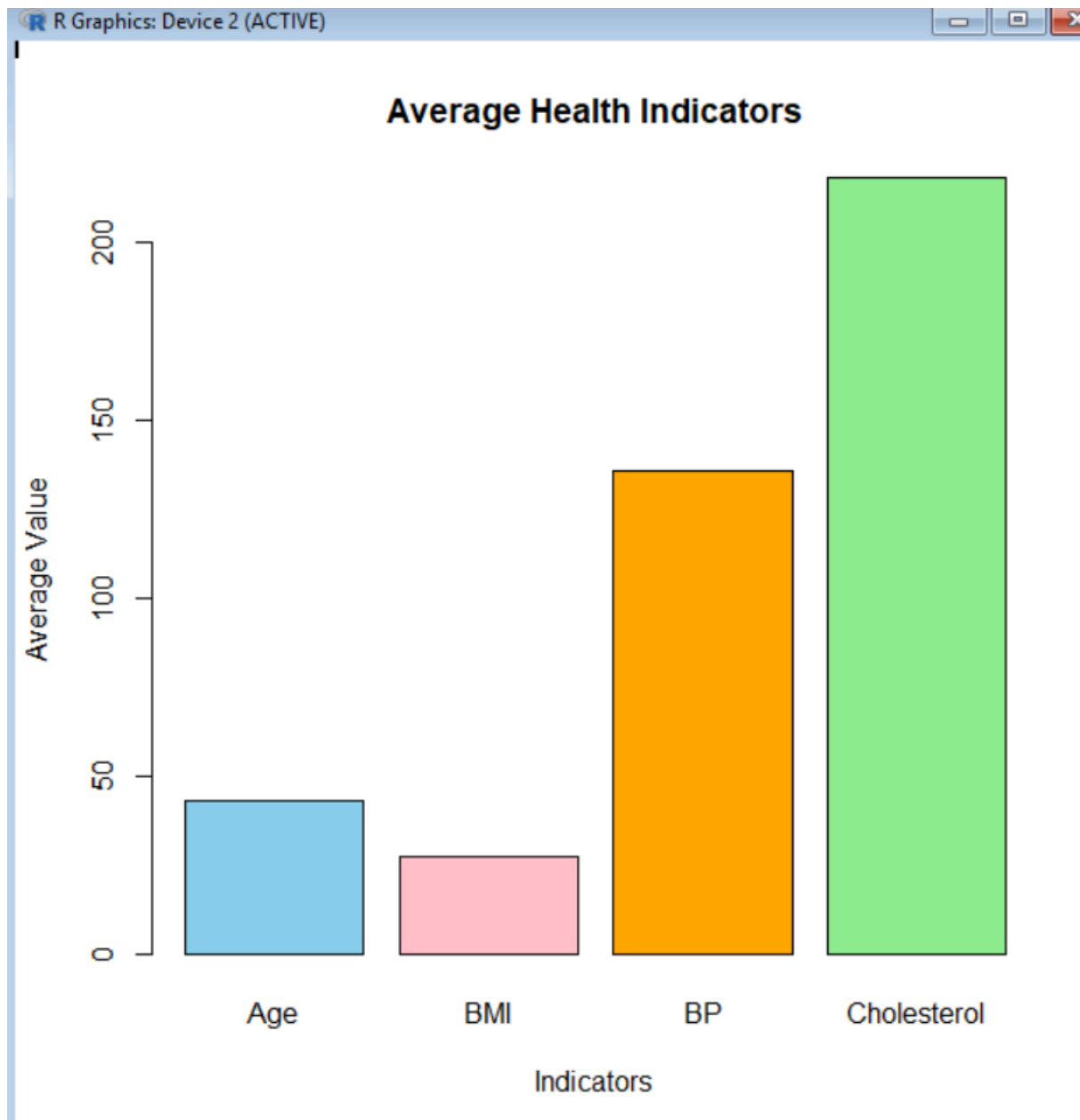
3)CODE

Bar Chart of Average Health Indicators

```
data <- read.csv("C:/Users/selco/OneDrive/Desktop/Patient_Health_Risk.csv")
```

```
avg <- colMeans(data[,2:5])
```

```
barplot(avg,  
  col=c("skyblue","pink","orange","lightgreen"),  
  main="Average Health Indicators",  
  xlab="Indicators",  
  ylab="Average Value")
```



Interpretation: Higher average BP and Cholesterol indicate an increased overall health risk in the dataset.

SET 28. Vehicle Performance Analysis

Dataset:

Vehicle_ID

Engine_Size

(L)

Horsepower

Fuel_Efficiency

(km/l)

Top_Speed

(km/h)

Safety_Rating

g

V1

1.5

110

18

180

4

V2

2.0

150

15

200

5

V3

3.0

250

12

250

5

V4

2.5

200

14

220

4

V5

1.8

130

17

190

3

1. Using R, develop a violin plot to compare the distribution of Fuel_Efficiency across different Safety_Rating categories. Examine the spread, central tendency, and variability within each group, and interpret what this indicates about efficiency patterns.
2. Using R, construct a scatter plot to explore the relationship between Horsepower and Top_Speed, applying color differentiation based on Engine_Size. Analyze the strength and direction of correlation and find observable performance trends.
3. Using R, create a correlation heatmap for all numerical variables in the dataset. Identify the variables most strongly associated with Top_Speed and their influence on overall vehicle performance.
4. In Tableau, design an interactive dashboard incorporating bar charts and scatter plots, along with filters for Engine_Size and Safety_Rating, to highlight vehicles that balance fuel efficiency and high performance. Provide analytical interpretation of the insights

derived from the dashboard.

SET 29 .Student Academic Performance

Dataset:

Student_I

D

Ag

e

Study_Hour

s

Attendance

(%)

Test_Scor

e

Participation_Scor

e

S1 19 12 90 85 8

S2 21 8 70 70 7

S3 20 15 95 92 9

S4 22 10 85 80 8

S5 23 7 60 65 6

1. Using R, create a stacked area chart showing Test_Score and Participation_Score across

students. Analyze trends.

2. Using R, generate a boxplot of Study_Hours grouped by Attendance quartiles.

Interpret

patterns. Include axis labels, title, and colors for each quartile. Analyze the boxplot to

identify median and interquartile range of study hours per attendance quartile and

any

outliers in study hours.

3. Using R, construct a density plot for Test_Score to evaluate score distribution and outliers.

4. In Tableau, design an interactive scatterplot with Study_Hours on X-axis,

Test_Score on

Y-axis, colored by Participation_Score. Include filters for Age.

SET 30

Dataset: Mobile_App_Usage.csv

User_ID Gender Age Screen_Time

(hrs) App_Usage_Count Data_Used

(GB) Satisfaction Usage_Date

U01 Male 20 4.5 18 2.4 3 2025-01-08

U02 Female 22 6.0 25 3.8 5 2025-01-08

U03 Male 19 3.2 12 1.6 3 2025-02-11

U04 Female 21 7.1 30 4.5 5 2025-02-11

U05 Male 23 2.8 10 1.2 2 2025-03-14

U06 Female 20 5.4 22 3.1 4 2025-03-14

1. Import the dataset in R. Plot a histogram and density plot for Screen_Time. Add suitable colors

and labels. Interpret the screen-time distribution.

2. Create a scatter plot using R programming between Data_Used and Screen_Time. Calculate the

correlation between them. Add a trend line. Explain the relationship.

3. Using R, find the average Satisfaction score for each Gender. Create a bar chart for the averages.

Add value labels. Interpret user satisfaction patterns.

4. Import Mobile_App_Usage.csv into Tableau. Create a bubble chart (Screen Time vs Data Used).

Use color for Gender and size for App Usage Count. Create a line chart for usage trends. Add

filters for Gender and Usage_Date. Design a mini dashboard.